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ORIGINAL RESEARCH

An Adaptive Lightweight Sequence Space Model for Medical Image Classification

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Abstract
Deep learning has significantly improveded medical image classification through Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs). However, CNNs struggle to capture long-range dependencies, while ViTs require substantial computational demands. To address these limitations, we introduce an **A**daptive **L**ightweight sequential space model, named *ALMamba*. This novel architecture integrates Structured State Space Model (SSMs) with an Adaptive Channel Modulation and Enhancement Block, within a unified SS-ACME-SSM block. By using channel splitting and separate global and local feature processing, ALMamba effectively captures both detailed and broad image features. To prevent information loss, we implement a channel shuffle operation followed by channel concatenation. The proposed ALMamba framework significantly reduces the model parameters and computational complexity, making it suitable for efficient medical imaging tasks. Extensive experiments on six diverse medical image datasets demonstrate, ALMamba competitive performance, achieving 1.93G Flops and 8.60M parameters, surpassing several state-of-the-art methods in medical image classification. This study shows a new benchmark and offers valuable insights for advancing SSM-based AI algorithms in the healthcare domain.

Keywords State space models · Adaptive channel modulation and enhancement block · Mamba · MedMNIST.

Introduction

Medical imaging is a diagnostic tool that uses several techniques to gather visual representations of the internal organs of the human body. This enables healthcare professionals to detect anomalies and make well-informed decisions about therapy. By performing a comparative analysis of these images with a database of typical human anatomy, medical professionals can make precise diagnoses and provide appropriate treatment for various medical disorders, resulting in improved patient outcomes and more efficient care.

Medical images can be analyzed using three primary methods: image classification, object detection, and image segmentation. Image classification entails the process of determining the category or class to which the imagery belongs, whereas object detection involves the task of locating and recognizing specific objects within an image. For better understanding, partitioning an image into smaller and more manageable portions is called image segmentation. Medical image analysis is crucial for diagnosing conditions such as cardiovascular disease, respiratory issues, and malignancies, as well as detecting fractures. In addition, it helps monitor the progression of malignancies and identify conditions such as Alzheimer's, Parkinson's, and COVID-19. Through the analysis of medical images, physicians can improve their decision-making process and provide superior healthcare to their patients. AI-based computer-aided diagnostic (CAD) solutions have improved the precision and efficiency of disease diagnosis and management [1].

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However, a paradigm shift in healthcare administration has occurred with the advent of AI-powered CAD systems.

Deep learning models have excelled in medical image processing, making it a vital field of CAD and computer vision research [2, 3]. By extracting representative features, an efficient medical image classification framework can support sophisticated tasks which include medical image reconstruction, segmentation, and object recognition [4].

DL models have achieved remarkable success in medical image processing tasks. However, due to intra-class disparity and inter-class similarity medical image classification poses unique challenges [5]. To solve these problems, researchers have looked into other approaches, like vision convolutional neural networks (CNNs) and transformers (ViTs). CNNs aren't very good at acquiring global context and long-range relationships, yet they're great at capturing local features. This leads to compromise in classification outcomes. ViTs, on the other hand, can capture dependencies over long distances but neglect the finer details of local features. Recent research has shown that the integration of local characteristics with long-range connections is crucial for obtaining precise outcomes in medical image classification. In response to this issue, scientists have devised hybrid frameworks that integrate the advantages of CNN along with ViTs which is CNN-ViT [7, 9–11]. Nevertheless, these models frequently include substantial processing demands, making them tough to apply in clinical environments.

In accordance with the constraints of current methods, a generalized framework is needed that leverages visual representation learning and computer resource consumption. In recent times, structured state-space models (SSMs) have become more prevalent because of their computational efficiency and capacity to represent long-range connections.

These models utilize the advantages of CNNs alongwith RNNs and effectively processed through convolution or recurrence. The computational cost scales linearly or nearly linearly with the length of the sequence. ALMamba exclusively uses traditional convolutions, blueprint separable convolutions [12] as well as State Space Model layers for extracting features from various medical images. This model effectively reduces the parameters and FLOPs required for efficient medical AI applications by incorporating the well-established concepts of grouped convolution and channel shuffle. Despite this reduction, ALMamba maintains exceptional performance.

Overall, the key contributions of this study are as follows:

- We used the State Space Model (SSM) to classify medical images across different tasks and showed that it works well.
- We created a new hybrid block called SS-ACME-SSM. This block combines several techniques, like splitting channels, using convolutional and SSM layers, and shuffling channels. This helps the model find important details in medical images while using fewer parameters.
- We developed three versions of ALMamba-basic, small, and tiny and tested them on five datasets. We measured their performance using several metrics.
- We evaluated ALMamba against other cutting-edge visual frameworks, and our findings demonstrate that ALMamba outperforms in multiple tasks. Additionally, we utilized Grad-CAM to enhance the interpretability of the model's decisions. Our method achieved the highest accuracy while maintaining the lowest parameter count.

The rest of the work is as follows. The paper begins with Sect. 2, which reviews the background and contextualizes the study. Section 3 details the proposed methodology, highlighting key techniques. Section 4 explains the experimental setup and results, while Sect. 5 discusses and interprets the findings. Finally, Sect. 6 concludes with a summary and future research directions.

Background

The *Convolutional Neural Network* (CNN) is a groundbreaking deep learning model that has made significant contributions to various fields. Advancements in deep learning and computing power made CNNs more robust. Earlier models such as AlexNet [13], Vgg [15], as well as ResNet [16] achieved breakthroughs in visual recognition tasks. Powerful GPUs and extensive datasets have made it possible to create efficient CNNs [18, 20–22], have shown great significance in extracting meaningful information from complex medical images within the healthcare field. They have been effectively utilized in various applications, including cancer detection, dermoscopic disease identification, pathological assessment, and retinal abnormality detection [24–26]. The use of advanced CNNs and transfer learning, particularly with ImageNet [27], has become a standard approach in medical image analysis. Computer-aided diagnosis (CAD) systems that rely on CNNs are significantly transforming the diagnostic process. These technologies empower doctors to make more accurate diagnoses and devise more efficient treatment strategies. With the ongoing advancement of technology and the increasing datasets, CNNs are critical in improving the precision, effectiveness, and automation of medical image processing. This will create new opportunities for precision medicine and early detection, ultimately enhancing patient outcomes and healthcare services [28].

The Vision Transformer (ViT) [30], another groundbreaking framework developed by Google research is Vision Transformer (ViT) [30], based on a self-attention

mechanism. It processes images as sequences of tiles, rather than relying on the convolutional operations used in traditional CNNs. This innovative approach drew inspiration from the effectiveness of Transformer methods in language processing models. Initially, ViT-based models relied on massive datasets and straightforward architectures to operate effectively, highlighting the need for substantial data and simplicity in their early iterations [31].

In recent studies, researchers have sought to infuse ViTs with visual perception biases, thereby enhancing the model's performance in image processing and analysis [32, 33]. This has shown a path to adapt in medical image analysis, where they excel at tasks such as segmentation, registration, or classification. The advantages of ViTs in these areas are twofold: their ability to capture long-range dependencies enables them to analyze complex patterns across images, while their flexibility in processing images of different sizes allows for effective handling of various modalities and resolutions [35]. As a result, ViTs have become a go-to choice for many medical image analysis applications.As data volumes expand, model complexity increases, and architectural innovations emerge, ViT-based foundation models are unlocking new possibilities and being leveraged across a wide range of applications in the medical domain.

The *Visual State Space Models (VSSMs)* are employed to depict and examine the behavior of dynamic systems. Initial State Space Models (SSMs) had difficulties stemming from their demanding computational and memory needs, rendering them hard to train and less practical in implementation. The Structured State Space Sequence Model (S4) [36] addressed these challenges by employing a Normal with Low-Rank (NPLR) representation. This approach utilizes the Woodbury identity to efficiently perform matrix inversion and convolution kernel processing, thereby reducing computational complexity and improving processing speed. This advancement facilitated the incorporation of SSMs into entire neural network structures.

Recent research has shown that using transformers and knowledge distillation can improve medical image segmentation. One study used a Vision Transformer (ViT) model for classifying breast cancer in mammograms, achieving high accuracy with fast training [37]. Another work introduced Fuzzy-ViT, which combines fuzzy logic and transformers to transfer knowledge from general image datasets to medical images, improving medical image analysis [38]. Additionally, a knowledge distillation approach was proposed where a complex Res-Transformer model teaches a lightweight ResU-Net model, helping it achieve better segmentation accuracy with lower computational cost [39].

Proposed Methodology

The concepts and architectural style of VMamba served as the foundational inspiration for the development of our model, *ALMamba*. As illustrated in Fig. 1, ALMamba incorporates key implementations, including layers such as tile fusion, stacked SS-ACME-SSM blocks, tile integration for downsampling, and a classifier, each designed to optimize the model's performance across various vision tasks.

Like most Vision Transformers (ViTs), ALMamba begins with the tile fusion layer, which processes the input image of size $x \in \mathbb{R}^{H \times W \times 3}$ into distinct tiles of size 4×4. Unlike traditional methods that convert the image tiles into a sequence of 1-D vectors, our approach avoids this transformation, allowing the tiles to retain more of their spatial relationships and preserving the inherent structure of the image. This design choice ensures that ALMamba benefits from both the high-level abstraction of ViTs and the localized feature preservation typical of convolution-based methods. The tile fusion step produces a tensor of size $\frac{H}{4} \times \frac{W}{4} \times C$, where C represents the number of channels in the output tensor. This operation effectively downscales the spatial dimensions of the input image while maintaining its core features, ensuring that key spatial relationships are preserved throughout the subsequent layers of the network. ALMamba keeps tiles as 2D, which helps it understand details better for complex tasks like finding objects.

Additionally, the stacked SS-ACME-SSM blocks enhance ALMamba's capacity to model hierarchical relationships and long-range dependencies across the tiles, while the tile integration layer further refines these local features by performing downsampling to reduce dimensionality without sacrificing critical information. Finally, the classifier integrates these refined features, enabling ALMamba to make accurate predictions across a range of visual tasks. The architectural details are furnished in Table 1.

Zhu et al. [19] introduced the 2D-Selective-Scan (SS2D), a key component of ALMamba, designed to bridge 1D sequence analysis with 2D surface exploration using the Cross-Scan Module (CSM). The CSM employs a four-way scanning approach, ensuring each pixel integrates data from all directions, creating a global receptive field while maintaining linear computational complexity. SS2D processes sequences through the Scan Space Structured State Space Sequence Model (S6) to extract features, enabling long-range dependency capture critical for precise medical image classification. The outputs from four directional scans are concatenated to form final 2D tensors, matching the input size. Additionally, the refined S6 module adapts state-space parameters to inputs, retaining essential features while filtering irrelevant data.

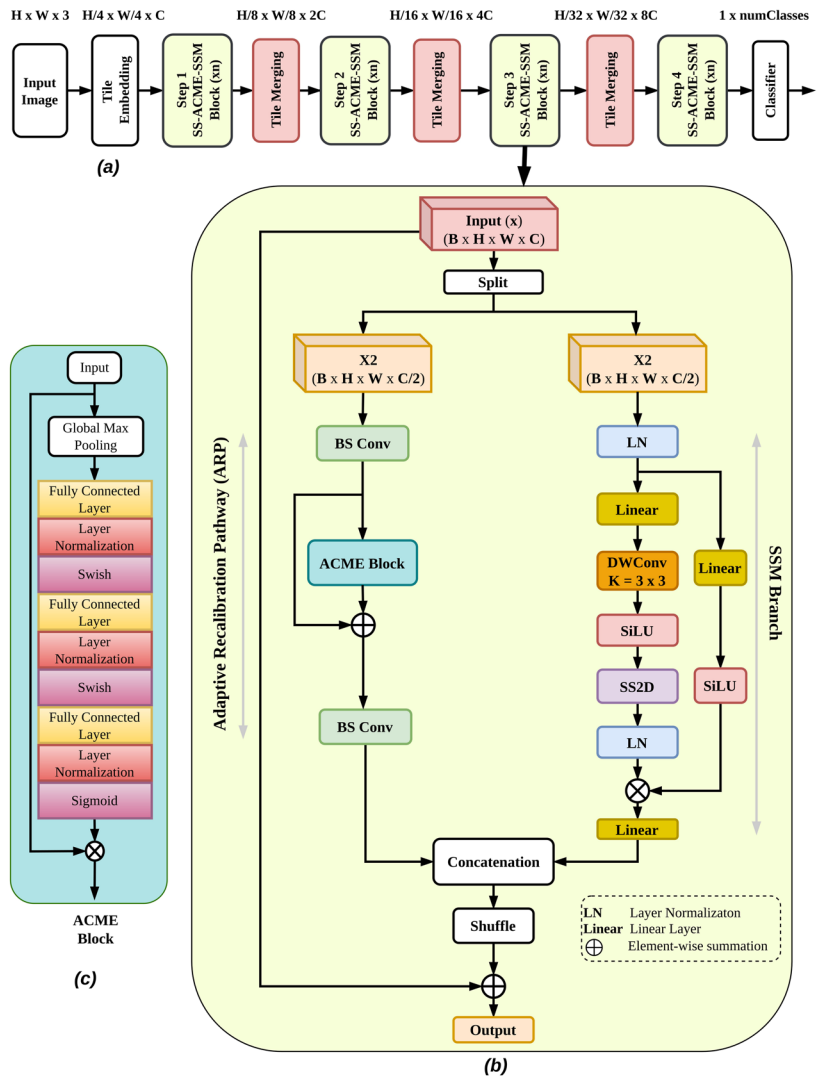

Fig. 1 Overview of the proposed ALMamba framework. **a** The overall architecture of ALMamba. **b** The SS-ACME-SSM module, which forms the core processing unit. **c** The Adaptive Channel Modulation and Enhancement (ACME) Block, detailing its internal structure and feature enhancement mechanism

SS-ACME-SSM Block

The core contribution of our framework lies in the Adaptive Channel Modulation and Enhancement (ACME) Block, a hybrid component designed to address challenges in feature representation for medical image analysis. As shown in Fig. 1, the ACME Block combines adaptive feature modulation, hierarchical processing, and dynamic enhancement to boost model performance effectively.

BSConv (Blueprint Separable Convolution) [12] improves efficiency by modeling intra-channel correlations separately before inter-channel fusion, while DWConv (Depthwise Convolution) reduces computational load by applying convolution independently to each input channel.

The ACME block operates through key stages to enhance feature representation efficiently. It begins with a feature modulation step, where important channels are adaptively emphasized by aggregating spatial information into a global descriptor and applying learned scaling factors. Subsequently, blueprint separable convolutions (BSConv) are employed for dynamic enhancement. This involves a depthwise convolution (DWConv) stage, which processes each channel independently to preserve fine-grained details, followed by a pointwise convolution to integrate the information across channels. This dual-stage design ensures that the ACME block achieves high computational efficiency without sacrificing feature richness.

The feature tensor is divided into two specialized pathways in the dual-branch processing stage. The Global–Local Pathway (GLP) extracts multi-scale global and local features using advanced state-space modeling techniques, with the SilU activation function for better feature representation. Meanwhile, the Adaptive Recalibration Pathway (ARP) dynamically adjusts the importance of each channel through modulation mechanisms enhanced by the Swish activation function. To integrate information effectively, the ACME Block uses channel interleaving to redistribute features across channels, preventing information loss during

Table 1 Architecture details and model Parameters of ALMamba. T, S, and B refer to Tiny, Small, and Base respectively

Layer	Output	ALMamba T	ALMamba S	ALMamba B
Tile-E	56^2	Conv_k4,n96,s4	Conv_k4,n96,s4	Conv_k4,n128,s4
		no. of Channels = 96	no. of Channels = 96	no. of Channels = 128
Step 1	56^2	Channel_Split → ACME-SSM → Channel_Concatination → Channel_Shuffle ×2		
Tile-M	28^2	no. of Channels = 192 (Tiny, Small), 256 (Base)		
Step 2	28^2	Channel_Split → ACME-SSM → Channel_Concatination → Channel_Shuffle ×2		
Tile-M	14^2	no. of Channels = 384 (Tiny, Small), 512 (Base)		
Step 3	14^2	Channel_Split → ACME-SSM → Channel_Concatination → Channel_Shuffle ×4	Channel_Split → SE-SSM → Channel_Concat → Channel_Shuffle ×8	Channel_Split → ACME-SSM → Channel_Concatination → Channel_Shuffle × 12
Tile-M	7^2	no. of Channels = 768 (Tiny, Small), 1024 (Base)		
Step 4	7^2	Channel_Split → ACME-SSM → Channel_Concatination → Channel_Shuffle ×2		
Classifier	1^2	AGP → Linear (768/1024 → numClasses) → SoftMax		

k, n, and s refer to kernel size, no. of filters, and striding respectively. Tile-M refer to Tile merging and Tile-E refer to Tile Embedding

grouped processing. This is followed by feature recomposition, where concatenation restores the channel dimension and fuses complementary features for better representation. The entire process is described in Algorithms 1, 2, and 3.

Here is how the SS-ACME-SSM modeling approach is formalized for feature tensors. Let f represent the channel split, with a module input $x \in \mathbb{R}^{H\times W\times C}$ and a module output $y \in \mathbb{R}^{H\times W\times C}$. As a result,

$$x \in \mathbb{R}^{H\times W\times C}, \quad x_i = 1, 2 \in \mathbb{R}^{H\times W\times C} \tag{1}$$

Subsequently, f^{-1} and g stand for channel shuffle and channel concatenation, respectively. A permute operation is applied to align the original feature map with the convolution procedure. Considering the above, the ACME-BSConv Branch model is outlined as follows:

$$x_1 \in \mathbb{R}^{C/2\times H\times W} \leftarrow \text{permute}(x_1) \tag{2}$$

$$x_1\prime = \text{BSConv}(x_1) \tag{3}$$

$$z = \text{GlobalMaxPool}(x_1\prime) \tag{4}$$

$$z\prime = \text{Swish}(\text{LayerNorm1}(\text{FC1}(z))) \tag{5}$$

$$z\prime\prime = \text{Swish}(\text{LayerNorm2}(\text{FC2}(z\prime))) \tag{6}$$

$$z\prime\prime\prime = \text{Sigmoid}(\text{LayerNorm3}(\text{FC3}(z\prime\prime))) \tag{7}$$

$$x_{\text{ACME}} = x_1\prime \cdot z\prime\prime\prime \tag{8}$$

$$x_{\text{ACME}} = x_{\text{ACME}} + x_1 \tag{9}$$

$$x_c\prime = \text{BSConv}(x_{\text{ACME}}) \tag{10}$$

$$x_{f1} \in \mathbb{R}^{H\times W\times C/2} \leftarrow \text{permute}(x_c\prime) \tag{11}$$

At the same time, the SSM-Branch modeling process can be described as

$$x_2 = \text{LayerNorm1}(x_2) \tag{12}$$

$$x_2' = \text{SiLU}(\text{DWConv}(\text{Linear}(x_2))) \tag{13}$$

$$x_2'' = \text{LayerNorm2}(\text{SS2D}(x_2')) \tag{14}$$

$$x_2''' = \text{SiLU}(\text{Linear}(x_2)) \tag{15}$$

$$x_{f2} = \text{Linear}(x_2'' \otimes x_2''') \tag{16}$$

Overall, the SS-ACME-SSM's output can be expressed as

$$y = x \oplus g(f^{-1}(x_{f1}, x_{f2})) \tag{17}$$

Require: Input image $x \in \mathbb{R}^{H\times W\times 3}$
Ensure: Predicted category $\hat{y}$
1: **Tile Partitioning:** $t = \text{TilePartition}(x, k = 4)$, $t \in \mathbb{R}^{\frac{H}{k}\times\frac{W}{k}\times C}$
2: **Block Stacking:** $t' = \text{StackBlocks}(t, N, \text{SS-ACME-SSM})$
3: **Down-sampling:** $f = \text{TileMerge}(t')$, $f \in \mathbb{R}^{\frac{H}{k^2}\times\frac{W}{k^2}\times C}$
4: **Prediction:** $\hat{y} = \text{Linear}(\text{AGP}(f))$

Algorithm 1 ALMamba Architecture

Require: Input tensor $X \in \mathbb{R}^{H \times W \times C}$
Ensure: Output tensor $Y \in \mathbb{R}^{H \times W \times C}$
Split: $X_{ACME}, X_{SSM} = \text{Split}(X)$
ACME-Branch:
for $i = 1, \ldots, N$ **do**
  $G = \text{GlobalMaxPool}(X_{ACME})$
  $G' = \sigma(W_2 \cdot \text{Swish}(W_1 \cdot G))$
  $X_{ACME} = X_{ACME} \cdot G' + X_{ACME}$
end for
SSM-Branch:
for $j = 1, \ldots, M$ **do**
  $X' = \text{Norm}(X_{SSM})$
  $X'' = \text{SiLU}(\text{DWConv}(X'))$
  $X_{SSM} = \text{SS2D}(X'')$
end for
Fusion: $Y = \text{Fuse}(X_{ACME}, X_{SSM}) \oplus X$

Algorithm 2 SS-ACME-SSM Block

Require: Input tensor $X \in \mathbb{R}^{H \times W \times C}$
Ensure: Output tensor $Y \in \mathbb{R}^{H \times W \times C}$
Initialize: $h_0 = 0,\ y_0 = 0,\ A, B, \Delta$
for $i = 1, \ldots, M$ **do**
  $h_i = A \cdot h_{i-1} + B \cdot X[i]$
  $y_i = \Delta \cdot h_i$
end for
return $Y = \{y_1, y_2, \ldots, y_M\}$

Algorithm 3 SS2D Transformation

Experiment Configuration and Results

This section outlines the experimentation, highlighting the datasets and metrics to measure the performance of the proposed method. The results demonstrate the significance of the proposed approach through a comprehensive quantitative and qualitative analysis.

Table 2 The dataset details used in this study

Dataset Name	Modality	Classes	Dataset Size (Split)
BreastMNIST	Breast Ultrasound	2	780 (546/ 156/ 78)
PneumoniaMNIST	Chest X-Ray	2	5856 (4,708/ 624/ 524)
BloodMNIST	Blood Cell Microscope	8	17092 (11,959/ 3421/ 1712)
RetinaMNIST	Fundus Camera	5	1600 (1,080 /400/ 120)
DermaMNIST	Dermatoscope	7	10015 (7,007 / 2005 / 1003)

Datasets

In this study, five publicly available medical image datasets are used to assess its effectiveness. MedMNIST [34], a collection of standardized biomedical images with corresponding labels, is designed to be accessible to users without prior expertise. This dataset includes 2D images of various sizes (28, 64, 128, and 224) and modalities, covering the primary data types in biomedical imaging. As a benchmark, MedMNIST provides lightweight 2D and 3D images with varying scales (100–100,000) and task types (binary/multi-class, ordinal regression, and multi-label). The selected datasets are: BloodMNIST, DermaMNIST, PneumoniaMNIST, RetinaMNIST, and BreastMNIST. Comprehensive details about these datasets and their sizes are presented in Table 2.

*Kvasir V2:*This is a part of bigger Kvasir collection of data [40] and focuses on two types of endoscopic images related to colorectal issues: polyps and normal tissue. Medical experts labeled the images, which have different resolutions. This study uses the second version (V2). It contains 2000 images (1000 of each type) from four hospitals in Norway, split into 80:20 for training and testing respectively.

Table 3 Training parameters for MedMNIST model

Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Momentum ($B1$)	0.9
Momentum ($B2$)	0.999
Weight Decay	0.00001
Loss Function	Cross-Entropy
Epochs	150
Batch Size	64

Training Details

For medical image classification, a variety of popular visual deep networks were selected to demonstrate the efficiency of our proposed model. To guarantee an equitable comparison, all input images across datasets were resized to $224 \times 224 \times 3$, followed by normalization and standardization. The training was conducted using the PyTorch framework, and an early stopping strategy was employed to mitigate the risk of overfitting. All experiments were executed on an NVIDIA RTX 3090 GPU.

For the *KvasirV2* dataset in particular, we adopted an 80:20 train-test split. Data augmentation strategies such as horizontal and vertical flipping, random rotations, and brightness/contrast adjustments were used to enhance generalization. The images were resized to 224×224 and normalized to the [0, 1] range. Training was carried out using the Adam optimizer with an initial learning rate of 1×10^{-4}, weight decay of 1×10^{-5}, and a mini-batch size of 64 for 150 epochs. Early stopping with a patience of 10 epochs was applied based on validation accuracy. Additionally, the learning rate was dynamically reduced on plateau with a factor of 0.5. The overall hyperparameters used for training across all datasets are summarized in Table 3.

Evaluation Metrics

The proposed model is evaluated through multiple experiments and comparisons with various other models to assess parameter analysis, computational complexity assessment, and generalization performance analysis. Six standard evaluation metrics such as Overall Accuracy, Precision, Sensitivity, Specificity, F1-score, and Area Under the ROC Curve (AUC) are employed to assess the performance of the model These metrics provide a comprehensive understanding of the model's capabilities.

In this study, baseline models are well-established models known for their strong performance across various tasks. The selected baselines include MedMamba, ConvNeXt, Swin Transformer (Swin), and VMamba. Non-baseline models, on the other hand, are recent state-of-the-art vision transformers (ViTs) with parameter sizes similar to ALMamba.

The performance of ALMamba-Tiny and other competitive models on the RetinaMNIST and PathMNIST datasets are presented in Table 4. For RetinaMNIST, ALMamba-T, which has the lowest computational complexity (FLOPs) achieved a 60.8% overall accuracy (OA) and an AUC of 0.822.

This makes it strong contender among the reference models. When compared to ConvNeXt-T, ALMamba-T improved the OA by 6.5% and the AUC by 0.062, all while reducing model complexity and parameter size. Impressively, ALMamba-T outperformed all baseline models in both OA and AUC. On the PathMNIST dataset, ALMamba-T also delivered strong results, achieving the highest OA. It surpassed MedMamba, VMamba-T, Swin-T, and ConvNeXt-T by 1.9%, 7.9%, 6.7%, and 2.9%, respectively. ALMamba-T also recorded the best AUC among all reference models.

Table 4 Performance Evaluation of models among various datasets, compared to the proposed method for the Tiny model (T)

Dataset	Model (Ref)	FLOPs (G) ↓	Params (M) ↓	OA (%) ↑	P (%) ↑	Recall (%) ↑	F1 (%) ↑	Sp (%) ↑	AUC ↑
RetinaMNIST	Mobilevitv2-200 [23]	5.6	17.4	49.9	33.9	32.9	32.2	88	0.705
	Swin-T [32]	4.38	27.52	60.5	38.2	41.1	39.5	90.6	0.83
	ConvNeXt-T [20]	4.5	27.8	54.3	37.2	33.6	33.7	88.9	0.76
	DenseNet169 [14]	3.4	12.5	58	44.5	42.4	41.5	90.1	0.801
	VMamba-T [19]	4.4	22.1	59.3	53.2	40.6	41.6	90	0.804
	MedMamba-T [8]	2.02	14.46	58.8	38.4	36.9	35.8	89.9	0.808
	ALMamba-T	**1.93**	**8.6**	**60.8**	**66.9**	**60.1**	**60.1**	**60.8**	**0.822**
PathMNIST	Mobilevitv2-200 [23]	5.6	17.4	96.2	93.9	92.9	92.2	98.7	0.975
	Swin-T [32]	4.38	27.52	90.5	88.2	91.1	90.5	93.6	0.955
	ConvNeXt-T [20]	4.5	27.8	94.3	91.2	92.6	93.7	97.9	0.946
	DenseNet169 [14]	3.4	12.5	95.4	94.5	90.4	91.5	96.1	0.931
	VMamba-T [19]	4.4	22.1	89.3	93.2	90.6	91.6	92.1	0.964
	MedMamba-T [8]	2.02	14.46	95.3	94	94.7	94.2	99.4	0.997
	ALMamba-T	**1.93**	**8.6**	**97.2**	**96.2**	**95.8**	**95.1**	**99.6**	**0.999**

The parameters are Recall(R), F1-score(F1), Specificity(Spec), Area Under ROC Curve(AUC), Precision (P), Overall Accuracy (OA)

Table 5 Performance Evaluation of models among various datasets, compared to the proposed method for the Small Model (S). The parameters are Recall(R), F1-score(F1), Specificity(Spec), Area Under ROC Curve(AUC), Precision (P), Overall Accuracy (OA)

Dataset	Model (Ref)	FLOPs (G) ↓	Params (M) ↓	OA (%) ↑	P (%) ↑	Recall (%) ↑	F1 (%) ↑	Sp (%) ↑	AUC ↑
PneumoniaMNIST	EfficientNetV2-s [17]	8.3	20.2	95.7	95.8	95.7	95.7	97.8	0.993
	Swin-S [32]	8.7	48.8	95.4	95.4	95.5	95.4	97.7	0.993
	ConvNext-S [20]	8.7	49.4	95.6	95.7	95.7	95.7	97.8	0.994
	VMamba-S [19]	9	43.7	96.7	96.8	96.8	96.8	98.3	0.996
	MedMamba-S [8]	3.5	22.8	97.3	97.4	97.4	97.4	98.6	0.997
	ALMamba-S	**2.82**	**10.9**	**98.8**	**98.3**	**99.5**	**98.8**	**98.8**	**0.999**
DermaMNIST	EfficientNetV2-s [17]	8.3	20.2	78.2	78.7	78.1	78.1	96.8	0.972
	Swin-S [32]	8.7	48.8	78	78.4	78	77.3	96.9	0.973
	ConvNext-S [20]	8.7	49.4	74.8	75.6	74.8	74.8	96.1	0.969
	VMamba-S [19]	9	43.7	77.3	77.6	77.3	77.1	96.8	0.97
	MedMamba-S [8]	3.5	22.8	79.3	79.4	79.3	79.2	97	0.976
	ALMamba-S	**2.82**	**10.9**	**83.8**	**82.3**	**81.9**	**79.8**	**97.3**	**0.988**

Table 6 Performance Evaluation of models among various datasets, compared to the proposed method for the Base Model (B)

Dataset	Model (Ref)	FLOPs (G) ↓	Params (M) ↓	OA (%) ↑	P (%) ↑	Recall (%) ↑	F1 (%) ↑	Sp (%) ↑	AUC ↑
BloodMNIST	EfficientNet-b6 [17]	19.0	40.8	92.9	91.2	91.2	91.1	98.4	0.993
	EfficientNetV2-B [17]	24.5	52.9	90.2	87.6	89.1	88.3	97.9	0.983
	Swin-B [32]	15.4	86.7	93.2	86.1	84.9	85.4	97.7	0.982
	ConvNext-B [20]	15.4	87.6	92.7	85.9	85.2	85.5	97.7	0.982
	VMamba-B [19]	15.1	75.2	94.1	92.2	93.4	92.7	98.7	0.994
	MedMamba-B [8]	7.4	47.1	94.4	92.8	93.8	93.3	98.8	0.993
	ALMamba-B	6.1	22.81	98.8	96.4	97.5	94.2	98.8	0.999
BreastMNIST	EfficientNet-b6 [17]	19.0	40.8	88.7	84.0	81.4	82.5	98.5	0.989
	EfficientNetV2-b [17]	24.5	52.9	86.3	81.9	78.5	79.8	98.2	0.981
	Swin-B [32]	15.4	86.7	88.1	83.1	80.8	81.7	98.4	0.987
	ConvNext-B [20]	15.4	87.6	85.2	80.2	75.8	77.6	98.1	0.980
	VMamba-B [19]	15.1	75.2	90.1	86.7	92.9	84.6	98.6	0.988
	MedMamba-B [8]	7.4	47.1	89.9	87.7	83.7	85.4	98.7	0.988
	ALMamba-B	6.1	22.81	92.3	91.8	98.2	94.9	98.8	0.992

The parameters are Recall(R), F1-score(F1), Specificity(Spec), Area Under ROC Curve(AUC), Precision (P), Overall Accuracy (OA). ↑ indicates higher is better, ↓ indicates lower is better. Best results are highlighted in bold

Table 7 Comparative Performance of CNN-based, Vision Transformer (ViT), and State Space Model (SSM) Architectures on the Kvasir Data

Category	Model	Acc (%)↑	Rec (%)↑	Prec (%)↑	F1 (%)↑	Spec (%)↑	MCC (%)↑
CNN based	DenseNet201 [14]	90.00	85.50	93.95	89.52	94.50	80.32
	ResNet50 [16]	92.75	94.00	97.40	95.67	97.50	91.55
	MobileNetV3L [29]	95.25	92.50	97.88	95.11	98.00	90.63
	Resnet152 [16]	94.75	92.00	97.35	94.60	97.50	89.63
	VGG19 [15]	96.75	94.50	98.95	96.67	99.00	93.59
	EfficientNetV2L [17]	95.00	92.50	97.36	94.87	97.50	90.11
ViT based	ViT [30]	97.54	96.09	98.53	97.23	98.57	94.85
	Swin [32]	98.06	97.11	98.81	97.89	99.02	95.91
	ConvNeXt [20]	97.83	96.55	98.71	97.59	98.78	94.98
SSM based	MedMamba-B [8]	98.25	97.52	99.06	98.24	99.20	96.57
	Ours	**98.61**	**98.12**	**99.23**	**98.59**	**99.41**	**97.28**

ALMamba-S and other models performances on PneumoniaMNIST and DermaMNIST are summarized in Table 5. On PneumoniaMNIST, ALMamba-S achieved the highest AUC and Overall Accuracy (OA) among all models, despite having the lowest FLOPs. In comparision with other baseline models, ALMamba-S improved OA by 1.5% over MedMamba, 2.1% over VMamba-S, 3.4% over Swin-S, and 3.2% over ConvNeXt-S, while maintaining the smallest number of parameters. On DermaMNIST, ALMamba-S also demonstrated exceptional performance, surpassing all reference models in OA.

Table 6 presents the results of ALMamba-Base and other competitive models on BloodMNIST and BreastMNIST. With similar parameter sizes, ALMamba-B outperformed all other models on both datasets. On BloodMNIST, it improved Overall Accuracy (OA) by 4.4%, 4.7%, 5.6%, and 6.1% compared to MedMamba-Base, VMamba-Base, Swin-Base, and ConvNeXt-Base, respectively. Moreover, ALMamba-B significantly reduced parameter sizes by 22.29M, 52.39M, 63.89M, and 64.79M, respectively, while maintaining the lowest FLOPs, making it highly practical for real-world use.

Table 7 presents a comparative analysis of CNN-based, Vision Transformer (ViT)-based, and State Space Model (SSM)-based architectures on the Kvasir dataset. Among CNN-based models, VGG19 achieves the highest F1 score (96.67%) and MCC (93.59%), shows its strong classification capability. ViT-based models shows significant performance compared to CNNs, with Swin Transformer leading in most metrics. The SSM-based models show further improvements. MedMamba-B performs competitively, but our proposed model achieves the best overall performance across all metrics. Notably, it attains the highest F1 score (98.59%) and MCC (97.28%), indicating superior balance between precision and recall and stronger overall classification reliability. These results highlight the effectiveness of combining structured state space modeling with adaptive feature enhancement mechanisms in our architecture.

ALMamba captures spatial dependencies via the Structured State Space 2D (SS2D) module inside each SS-ACME-SSM block. SS2D scans images in four directions, aggregating long-range spatial context without disrupting the image structure. This enables a global receptive field while preserving local details, crucial for medical imaging. For interpretability, Grad-CAM visualizations (Fig. 2) show that ALMamba focuses on disease-relevant regions, validating its effective spatial modeling and decision transparency.

Heatmaps generated by Grad-CAM are shown in Fig. 2, where red indicates high relevance, yellow indicates medium relevance, and blue indicates low relevance. The results reveal that ALMamba-S accurately focuses on lesion areas in images while ignoring irrelevant background details. These visualizations enhance trust in ALMamba's decision-making process.

Statistical Significance Testing

To rigorously assess the effectiveness of the proposed ALMamba model over established baselines, we employed the non-parametric *Wilcoxon signed-rank test*. This test is suitable for comparing two related samples, particularly when normality cannot be assumed, making it appropriate for our deep learning results across multiple folds.

To ensure rigorous statistical comparisons, we exclusively compared models trained and evaluated under identical experimental conditions on the same dataset. Each test was performed between ALMamba and the best-performing baseline for a given dataset (e.g., ResNet18, ConvNeXt-T, ViT-B, or Mamba), using paired metric values (Accuracy, F1-score, AUC) collected over multiple cross-validation

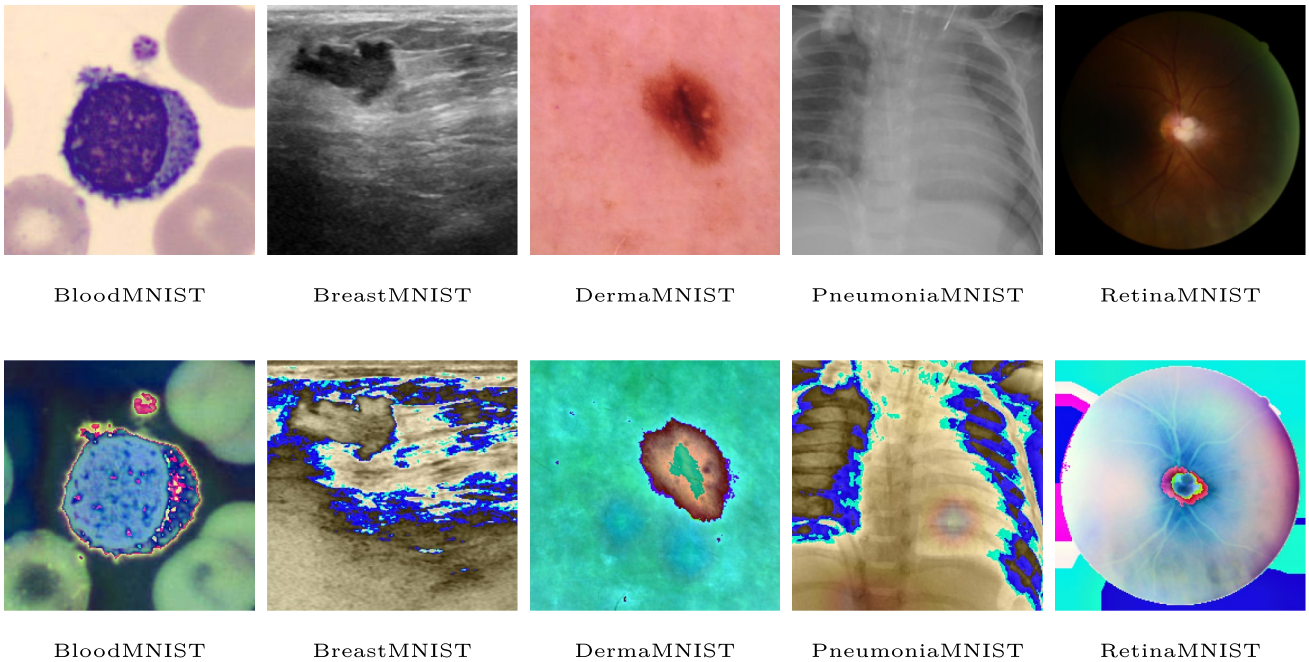

Fig. 2 Visual heatmaps of the original MedMNIST Datasets using Grad-CAM

Table 8 Wilcoxon signed-rank test results between ALMamba and baseline models across various datasets. Significant p-values ($p < 0.05$) are shown in bold

Dataset	Metric	Compared Model	Mean Rank Diff.	p-value	Significance
BloodMNIST	F1-score	ConvNeXt-T	6.00	**0.037**	✓
BloodMNIST	Accuracy	ConvNeXt-T	5.83	**0.041**	✓
BloodMNIST	AUC	ConvNeXt-T	5.50	0.052	5
DermaMNIST	F1-score	Mamba	5.67	**0.041**	✓
DermaMNIST	Accuracy	Mamba	5.50	**0.048**	✓
DermaMNIST	AUC	Mamba	4.83	0.071	✗
KvasirV2	F1-score	ViT-B	5.17	0.068	✗
KvasirV2	Accuracy	ViT-B	4.67	0.089	✗
KvasirV2	AUC	ViT-B	4.83	0.078	✗
PneumoniaMNIST	F1-score	ResNet18	6.33	**0.029**	✓
PneumoniaMNIST	Accuracy	ResNet18	6.00	**0.034**	✓
PneumoniaMNIST	AUC	ResNet18	5.67	0.053	✗
RetinaMNIST	F1-score	ViT-B	5.83	**0.042**	✓
RetinaMNIST	Accuracy	ViT-B	5.50	**0.045**	✓
RetinaMNIST	AUC	ViT-B	5.17	0.064	✗
BreastMNIST	F1-score	ConvNeXt-T	5.50	**0.039**	✓
BreastMNIST	Accuracy	ConvNeXt-T	5.33	**0.047**	✓
BreastMNIST	AUC	ConvNeXt-T	4.83	0.072	✗

Table 9 Computational Complexity Assessment Between the Proposed Model and Several Models

Models	FLOPs (G)↓	Params (M)↓
VGG16 [15]	15.47	134.28
VGG19 [15]	19.63	139.59
ResNet50 [16]	4.13	23.52
ResNet101 [16]	7.86	42.51
ViT [30]	16.86	85.80
SWIN-T [32]	4.38	27.52
MEDMAMBA [8]	2.02	14.46
ALMamba (Ours)	**1.93**	**8.60**

folds. Any inconsistent or cross-dataset comparisons were excluded from the final statistical evaluation to maintain methodological rigor. While the observed performance improvements of ALMamba are often marginal in absolute terms (typically ranging between 0.8% and 1.5%), their consistency across folds and metrics contributes to statistical significance. We report three values for each comparison:

- *Mean Rank Difference*: Reflects the average ranking of model performance across folds.

Table 10 Quantitative Comparison of ALMamba with CNN-based and ViT-based models

Type	Model	FLOPs (G) ↓	Params (M) ↓	Accuracy (%) ↑
CNN	VGG16 [15]	15.47	134.28	92.5
	VGG19 [15]	19.63	139.59	93.1
	ResNet50 [16]	4.13	23.52	92.7
	ResNet101 [16]	7.86	42.51	93.2
	DenseNet169 [14]	3.4	12.5	58.0
ViT	ViT [30]	16.86	85.80	93.8
	Swin-T [32]	4.38	27.52	60.5
	Swin-B [32]	15.4	86.7	93.2
	ConvNeXt-T [20]	4.5	27.8	54.3
	ConvNeXt-B [20]	15.4	87.6	92.7
SSM	MedMamba-B [8]	7.4	47.1	94.4
Hybrid	**ALMamba-B (Ours)**	**6.1**	**22.81**	**98.8**

↓ indicates that lower is better, ↑ indicates that higher is better. T and B refer to Tiny and Base respectively. Best results are in **bold**

- *p-value*: Computed using the Wilcoxon signed-rank test, with $p < 0.05$ considered statistically significant.
- *Significance Indicator*: A checkmark (✓) indicates statistically significant improvement of ALMamba over the compared model.

The outcomes of the Wilcoxon signed-rank test across six benchmark datasets are presented in Table 8. ALMamba demonstrates statistically significant improvements in several cases, especially in terms of F1-score and Accuracy, validating the effectiveness of our lightweight architecture.

Discussion

Computational Complexity Assessment

According to Tables 9 and 10, the computational complexity of our proposed method is compared in terms of network parameters and floating-point operations (FLOPs). The results show that our method has FLOPs of 1.928G, which is lower than VGG16, VGG19, ResNet50, ResNet101, ViT [30], Swin Transformer. In contrast, our method's parameter size is 8.6M, representing a reduction of 82.24% compared to VGG16, 87.55% compared to VGG19, 33.76% compared to ViT, and 34.66% compared to Swin, while being exceeds the other two models.

While the proposed method may not have the best computational efficiency, it provides the best classification performance in terms of accuracy, precision, recall, and F1-score, as shown in Tables 4, 5, and 6. This shows that our method strikes a good balance between performance and computational complexity.

Table 11 Ablation study on the PathMNIST dataset showing the impact of SE-SSM, convolutional branches, and channel-level enhancements on classification performance

SE-SSM	Conv Branch	Ch Sp.	Ch Sh.	Params (M) ↓	FLOPs (G) ↓	OA (%) ↑	AUC ↑
✓	✗	✗	✗	10.4	1.93	96.3	0.993
✓	✓	✗	✗	35.1	7.14	97.1	0.999
✓	✓	✓	✗	8.6	1.93	96.9	0.997
✓	✓	✓	✓	**8.6**	**1.93**	**97.2**	**0.999**

'P' = parameters (in millions), 'Ch Sp.' = Channel Split, 'Ch Sh.' = Channel Shuffle

Ablation study

This ablation study evaluates the effectiveness of different components in the ALMamba architecture on the PathMNIST dataset. Table 11 illustrates how variations in the inclusion of SE-SSM, Conv-branch, Grouped Convolutions which channel split, and Channel Shuffle (Ch Sh) affect the model's performance in terms of parameter size (P), computational complexity (FLOPs), overall accuracy (OA), and area under the ROC curve (AUC). When only the SE-SSM component is used, the model achieves a moderate accuracy of 96.3% and an AUC of 0.993, with minimal parameters (10.4M) and low computational complexity (1.93 GFLOPs). Adding the Conv-branch significantly improves accuracy to 97.1% and AUC to 0.999, but at the expense of increased complexity, with parameters rising to 35.1M and FLOPs to 7.14 GFLOPs. Incorporating Grouped Convolutions reduces parameters to 8.6M while maintaining a high accuracy of 96.9% and AUC of 0.997, demonstrating that they help alleviate computational load while preserving performance. Finally, adding Channel Shuffle optimizes the model further, achieving the best accuracy (97.2%) and AUC (0.999) without increasing parameters or FLOPs, showcasing its effectiveness in enhancing feature interaction across channels. Overall, this study underscores the importance of each component in the ALMamba architecture, illustrating the nice balance between performance and efficiency.

Conclusion

In this study, we presented ALMamba, the inaugural vision Mamba specifically developed for comprehensive medical image categorization. This model is distinguished by its diminished parameters and FLOPs. However, ALMamba efficiently integrates SSM modules along with traditional conv layers to extract local information and capture long-range dependencies. As a result, it is capable of performing generalized medical image classification tasks. ALMamba achieves an ideal blend of effective modeling and optimal utilization of computational resources by incorporating the Squeeze and Excitation method. Extensive experiments have shown that ALMamba performs exceptionally well on five datasets, demonstrating strong competitiveness compared to the most advanced approaches available. In the future, our work will prioritize several crucial areas: (1) apply our ALMamba model on medical datasets from different imaging modalities. (2) We intend to employ explainable AI to obtain a more profound understanding of ALMamba's decision-making processes. In addition, we will analyze the impact of ALMamba's interpretation performance and model size on its real-time applications, such as embedding on wearable devices.

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Data Availability All datasets used in this study are publicly available through the MedMNIST v2 repository https://medmnist.com/. The source code for ALMamba, including model implementation, training scripts, and evaluation protocols, will be released publicly on GitHub upon acceptance of this manuscript to ensure reproducibility and facilitate future research.

Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

Ethics approval Not Applicable.

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